

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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1. TITLE

Design of a Styled Webpage with CSS3 Flexbox Layout and HTML5 Canvas Graphics

2. AIM

To develop a modern styled webpage applying external CSS3 gradients, navigation bar hover effects, flexbox cards, and dynamically drawing 2D shapes on an HTML5 canvas using JavaScript.

3. OBJECTIVE

The objectives of this experiment are:

- To master external CSS3 stylesheet linking and global styling.
\n
- To apply linear gradients and text shadows to header elements.
\n
- To implement responsive flexbox layouts using display: flex and gap.
\n
- To understand the HTML5 <canvas> coordinate system and 2D rendering context.
\n
- To draw rectangles, arcs, lines, and text programmatically via JavaScript.

4. THEORY

CSS3 introduces advanced visual styling capabilities including multi-color linear gradients, box shadows, and Flexbox for effortless alignment and distribution of UI cards. The HTML5 <canvas> element provides a scriptable 2D drawing surface. Using JavaScript's getContext("2d"), developers can programmatically render vector shapes, paths, strokes, fills, and custom text onto the canvas.

5. ALGORITHM / PROCEDURE

1. Create index.html and link external stylesheet style.css.
\n
2. Design a gradient header using linear-gradient().
\n

3. Build a centered navigation bar with link hover color transitions.

\n

4. Declare an HTML5 canvas element with id='myCanvas'.

\n

5. Implement a .container with display: flex and gap containing styled cards.

\n

6. Write JavaScript to get the canvas 2D context using getContext('2d').

\n

7. Use fillRect() to draw a rectangle, arc() to draw a circle, moveTo()/lineTo() for a line.

\n

8. Test and verify the styled webpage and canvas graphics in a browser.

6. PROGRAM

index.html

```

<!DOCTYPE html>
<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>My Styled Webpage</title>

    <!-- External CSS -->
    <link rel="stylesheet" href="style.css">

    <!-- Google Font -->
    <link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;500;700&display=swap" rel="stylesheet">

</head>

<body>

    <!-- ===== HEADER ===== -->

    <header>

        <h1>ANITS Web Technology Showcase</h1>

        <p>
            HTML5 Graphics & CSS3 Modern Web Styling
        </p>

    </header>

    <!-- ===== NAVIGATION ===== -->

    <nav>

        <a href="#">🏠 Home</a>

        <a href="#canvas">🎨 Canvas</a>

        <a href="#cards">💻 Technologies</a>

        <a href="#footer">📞 Contact</a>

    </nav>

    <!-- ===== HERO SECTION ===== -->

    <section class="hero">

```

```

<div class="hero-text">

    <h2>Welcome to My Styled Webpage</h2>

    <p>

        This webpage demonstrates HTML5 Canvas and CSS3 styling
        techniques including gradients, shadows,
        hover effects and modern layouts.

    </p>

    <button>Explore More</button>

</div>

</section>

<!-- ===== CANVAS ===== -->

<section id="canvas">

    <h2 class="section-title">

        HTML5 Canvas Graphics

    </h2>

    <canvas id="myCanvas" width="850" height="350">

        Your browser does not support Canvas.

    </canvas>

</section>

<!-- ===== CARDS ===== -->

<section id="cards">

    <h2 class="section-title">

        Web Technologies

    </h2>

    <div class="card-container">

        <!-- Card 1 -->

        <div class="card">

```

```
<h2>🖥️ HTML5</h2>

<p>

    HTML5 provides semantic elements,
    multimedia support,
    forms,
    canvas
    and responsive webpage structure.

</p>

<button>Learn HTML5</button>

</div>

<!-- Card 2 -->

<div class="card">

    <h2>🎨 CSS3</h2>

    <p>

        CSS3 enhances webpages using gradients,
        shadows,
        transitions,
        animations,
        hover effects
        and modern layouts.

    </p>

    <button>Learn CSS3</button>

</div>

<!-- Card 3 -->

<div class="card">

    <h2>🖌️ Canvas</h2>

    <p>

        Canvas allows drawing
        rectangles,
        circles,
        lines,
        text,
        games,
        animations
        and graphics using JavaScript.
```

```

        </p>

        <button>Learn Canvas</button>

    </div>

</div>

</section>

<!-- ===== FOOTER ===== -->

<footer id="footer">

    <h3>

        Developed By

    </h3>

    <p>

        Gorapalli Sravya

    </p>

    <p>

        CSE (AI & ML)

    </p>

    <p>

        Anil Neerukonda Institute of Technology and Sciences (ANITS)

    </p>

    <br>

    <p>

        © 2026 All Rights Reserved

    </p>

</footer>

<!-- ===== CANVAS SCRIPT ===== -->

<script>

```

```
const canvas = document.getElementById("myCanvas");

const ctx = canvas.getContext("2d");

/* Rectangle */

ctx.fillStyle = "#2563eb";

ctx.fillRect(40, 40, 180, 120);

/* Circle */

ctx.beginPath();

ctx.arc(380, 100, 60, 0, Math.PI * 2);

ctx.fillStyle = "#10b981";

ctx.fill();

/* Line */

ctx.beginPath();

ctx.moveTo(520, 40);

ctx.lineTo(780, 160);

ctx.strokeStyle = "#ef4444";

ctx.lineWidth = 5;

ctx.stroke();

/* Text */

ctx.font = "30px Poppins";

ctx.fillStyle = "#111827";

ctx.fillText("Gorapalli Sravya", 240, 280);

</script>

</body>

</html>
```


style.css

```

*{
margin:0;
padding:0;
box-sizing:border-box;
}

body{
font-family:'Poppins',sans-serif;
background:linear-gradient(135deg,#8fb1de,#c9d2db,#56d6e2);
color:#333;
}

header{
background:linear-gradient(90deg,#567bd0,#2563eb);
color:white;
text-align:center;
padding:40px;
text-shadow:2px 2px 6px rgba(0,0,0,.4);
}

nav{
display:flex;
justify-content:center;
gap:30px;
background:#111827;
padding:18px;
position:sticky;
top:0;
}

nav a{
color:white;
text-decoration:none;
font-weight:600;
padding:10px 18px;
border-radius:30px;
transition:.4s;
}

nav a:hover{
background:#3b82f6;
color:#fff176;
transform:translateY(-3px);
}

.hero{
padding:35px 20px;
text-align:center;
}

.hero-content{
max-width:800px;
margin:auto;
}

```

```

background:rgba(255,255,255,.35);
backdrop-filter:blur(12px);
padding:40px;
border-radius:20px;
box-shadow:0 8px 20px rgba(0,0,0,.2);
}

.hero h2{
font-size:40px;
margin-bottom:15px;
color:#1e3a8a;
text-shadow:1px 1px 2px #bbb;
}

.hero p{
font-size:18px;
margin-bottom:25px;
}

button{
padding:12px 25px;
border:none;
border-radius:30px;
background:linear-gradient(90deg,#2563eb,#06b6d4);
color:white;
font-size:16px;
cursor:pointer;
transition:.4s;
box-shadow:0 5px 15px rgba(0,0,0,.3);
}

button:hover{
transform:scale(1.08);
background:linear-gradient(90deg,#16a34a,#22c55e);
}

.section-title{
text-align:center;
font-size:34px;
margin:40px 0 25px;
color:#1e3a8a;
}

#canvas{
text-align:center;
}

canvas{
background:white;
border:4px solid #2563eb;
border-radius:20px;
box-shadow:0 8px 20px rgba(0,0,0,.3);
}

```

```

.card-container{
display:flex;
justify-content:center;
flex-wrap:wrap;
gap:30px;
padding:20px;
}

.card{
width:300px;
background:rgba(255,255,255,.45);
backdrop-filter:blur(12px);
padding:25px;
border-radius:20px;
border:1px solid rgba(255,255,255,.4);
box-shadow:0 8px 20px rgba(0,0,0,.2);
transition:.4s;
text-align:center;
}

.card:hover{
transform:translateY(-10px) scale(1.03);
}

.card h3{
font-size:26px;
margin-bottom:15px;
color:#2563eb;
text-shadow:1px 1px 2px #ddd;
}

.card p{
margin-bottom:20px;
line-height:1.7;
}

footer{
background:#0f172a;
color:white;
text-align:center;
padding:18px;
margin-top:20px;
}

@media(max-width:900px){
nav{
flex-wrap:wrap;
gap:10px;
}
.card-container{
flex-direction:column;
align-items:center;
}
.hero h2{

```

```
font-size:30px;
}
canvas{
width:95%;
height:auto;
}
}
```

7. CODE EXPLANATION

Code / Syntax	Explanation
linear-gradient(...)	Generates a smooth color transition across the header background.
display: flex; gap: 30px;	Aligns child card elements in a flexible horizontal row.
canvas.getContext('2d')	Returns a drawing context on the canvas for 2D graphics rendering.
ctx.fillRect(x, y, w, h)	Draws a filled rectangle on the canvas at specified coordinates.
ctx.arc(x, y, r, 0, 2*PI)	Draws a full circular path.

8. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	CSS Linking	Load index.html	External CSS styles applied correctly	Pass
TC-02	Flexbox Cards	Inspect card container	Cards displayed side-by-side with shadows	Pass
TC-03	Canvas Shapes	Observe canvas element	Shapes rendered at exact coordinates	Pass

9. OUTPUT

Output 1: Design of a Styled Webpage with CSS3 Flexbox Layout and HTML5 Canvas Graphics

Screenshot not found: observation tasks/task 2\result.png

10. RESULT

Thus, the styled webpage with external CSS3 and interactive HTML5 Canvas graphics was successfully implemented and verified with expected visual rendering across all test cases.

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